

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda — Remote (UTC+3)

PROFESSIONAL SUMMARY

Backend developer with 6+ years of production **Node.js**, building and operating **REST APIs** with **Express.js** for international clients across the US, Singapore and East Africa. Led the end-to-end, zero-downtime migration of 20+ production applications from Parse/MongoDB to **Firestore/GCP**, building a real-time two-way MongoDB↔Firestore sync on Google Pub/Sub and removing \$5,000/month of infrastructure cost. Architected an event-driven microservices backend on Apache Kafka, **Docker** and **Kubernetes**, and ships backend features measured in business value — Express Shipping added \$40 revenue per order. Fully remote since 2020, UTC+3, working in agile teams reporting directly to the CTO.

CORE COMPETENCIES

Node.js

REST API Design

Express.js

SQL & NoSQL Databases

Docker & Kubernetes

GCP / Firestore / AWS

Microservices & Event-Driven Architecture

Zero-Downtime Migrations

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- Led the **end-to-end migration of 20+ applications** from Parse/MongoDB to Firestore/GCP with **zero downtime**, reporting directly to the CTO
- Implemented real-time two-way sync between MongoDB and Firestore using **Node.js** and **Google Pub/Sub** for message processing
- Migrated file storage from Amazon S3 to GCS via a Python script
- Saved the company **\$5,000/month** by migrating all services off AWS
- Implemented Express Shipping, generating an additional **\$40 revenue per order**
- Built a 3D visualisation feature for customers using video overlay and the ffmpeg library, increasing buyer conversion

MTailor

Mar 2022 – Jul 2022

Contractor — US Remote

- Hired via Upwork to execute a complex backend migration from Parse/MongoDB to Firestore and GCP
- Built a migration script to read data from MongoDB and write to Firestore after running complex processing logic
- Wrote documentation for fellow engineers on working with the new Firestore SDKs (Node.js and Python)
- Trained the Ops team on the new Firestore Dashboard and backup procedures

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Extended backend **REST APIs** using Firestore Cloud Functions and **Express.js** hosted on Heroku
- Built ad-hoc jobs to query **over 2 million Firestore records**, keeping customer prices current from Morningstar APIs
- Led optimisation of a PWA delivering real-time stock market prices from a Firestore backend
- Added responsive UI pages to the PWA using HTML, Tailwind and React

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for DeFi applications using Web3 and **Node.js** for a US agency with demanding clients, reporting directly to the CTO
- Delivered React UIs from Figma designs under tight timelines

CodeBits

Jan 2020 – Jul 2021

Software Engineer — Uganda Remote

- Architected a **microservices backend** for FIDA Uganda's case management app using Apache Kafka, **Docker** and **Kubernetes**
- Led a team of 4 developers** building legal tech systems for NGOs in Uganda
- Built a USSD service for legal aid providers communicating with a **Node.js** backend via gRPC
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda
- Developed a cross-platform mobile app for LASPNET using React Native and Expo (iOS & Android), with Firestore push notifications

PROJECTS

Parse → Firebase Migration Zero downtime

Migrated 20+ production applications off an end-of-life Parse/MongoDB stack onto Firebase/GCP in waves, with a rollback path at every stage and no customer-facing downtime. Removed significant recurring infrastructure cost.

Node.js · Firebase · Firestore · GCP · MongoDB · Python

Real-time Pub/Sub Sync Pipeline Data consistency

Built a live MongoDB ↔ Firestore two-way sync supporting parallel production traffic during migration, using idempotent handlers and origin tagging to prevent write loops and divergence.

Node.js · Google Pub/Sub · Firestore · MongoDB

FIDA Uganda Microservices Backend Architecture

Event-driven case management backend for a legal-aid NGO: service decomposition over Kafka and NATS Streaming, containerised and orchestrated, with a gRPC-backed USSD channel for low-connectivity users.

Apache Kafka · NATS Streaming · Docker · Kubernetes · gRPC · Node.js

Healthcare WhatsApp Chatbot Integrations

Conversational intake service integrating the WhatsApp Business API with OpenAI and Google Vision for prescription OCR.

Node.js · WhatsApp Business API · OpenAI · Google Vision

EDUCATION

BSc Computer Engineering — Makerere University, Kampala, Uganda

Aug 2018 – Sep 2022

SKILLS

Backend: Node.js, Express.js, REST APIs, gRPC, Apache Kafka, NATS Streaming, Python, Go **Databases:** SQL, MongoDB, Firestore

Cloud & DevOps: GCP (Firestore, Cloud Functions, Pub/Sub, GCS), Firebase, AWS (S3, EC2, EBS), Docker, Kubernetes, Linux, Git

Frontend: React, React Native, HTML, CSS, Tailwind **Also:** Rust (intermediate) **Languages:** English (fluent), Kiswahili